

A Proof of Conjecture 000000000050

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Claude (Opus 5)[†]

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Abstract

Conjecture 000000000050 asserts that every prime is the permanent of a square matrix with entries in $\{0, 1\}$. We prove it, and more: every natural number is, by an explicit $(n+1) \times (n+1)$ construction whose permanent is $n+1$. Primality plays no role. The proof is formalized in Lean 4 against Mathlib, using Mathlib's `Matrix.permanent`.

1 Statement

For a square matrix $A = (a_{rc})$ over a commutative semiring, indexed by a finite set I , the permanent is

$$\text{per } A = \sum_{\sigma \in \mathfrak{S}_I} \prod_{i \in I} a_{\sigma(i), i},$$

the determinant without signs. Conjecture 000000000050 asserts that for every prime p there is a square matrix with entries in $\{0, 1\}$ whose permanent is exactly p .

2 The construction

Definition 1. For $n \geq 0$ let W_n be the $(n+1) \times (n+1)$ matrix with rows and columns indexed by $\{0, 1, \dots, n\}$ and entries

$$(W_n)_{r,c} = \begin{cases} 1, & \text{if } r = 0 \text{ or } c = 0 \text{ or } c = r, \\ 0, & \text{otherwise.} \end{cases}$$

That is: the first row is all ones, and row $r \neq 0$ has ones exactly in column 0 and column r . For $n = 4$,

$$W_4 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Its entries lie in $\{0, 1\}$ by construction.

Lemma 2. A permutation σ of $\{0, \dots, n\}$ satisfies $\prod_i (W_n)_{\sigma(i), i} = 1$ if and only if

$$\sigma(i) \in \{0, i\} \quad \text{for every } i \neq 0, \tag{1}$$

and $\prod_i (W_n)_{\sigma(i), i} = 0$ otherwise.

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Proof. All entries are 0 or 1, so the product is 1 exactly when every factor is, and 0 otherwise. The factor at index i is $(W_n)_{\sigma(i),i}$, which is 1 precisely when $\sigma(i) = 0$ or $i = 0$ or $i = \sigma(i)$. Requiring this for every i is exactly (1), the index $i = 0$ imposing nothing. $\square$

Lemma 3. *The permutations satisfying (1) are exactly the transpositions $\tau_k = (0\ k)$ for $k \in \{0, \dots, n\}$, where τ_0 is the identity. They are pairwise distinct, so there are $n + 1$ of them.*

Proof. Each τ_k satisfies (1): for $i \neq 0$ we have $\tau_k(i) = 0$ if $i = k$, and $\tau_k(i) = i$ otherwise.

Conversely let σ satisfy (1) and put $k = \sigma(0)$; we claim $\sigma = \tau_k$. At 0 both sides give k . Let $i \neq 0$.

If $\sigma(i) = i$, then $i \neq k$: otherwise $\sigma(0) = k = i = \sigma(i)$ would give $0 = i$. Hence $\tau_k(i) = i = \sigma(i)$.

If instead $\sigma(i) = 0$, we show $i = k$. First $k \neq 0$, since $k = 0$ would give $\sigma(0) = 0 = \sigma(i)$ and so $i = 0$. Applying (1) at $k \neq 0$ gives $\sigma(k) \in \{0, k\}$; and $\sigma(k) = k$ is impossible, since with $\sigma(0) = k$ it would force $k = 0$. So $\sigma(k) = 0 = \sigma(i)$, whence $i = k$ and $\tau_k(i) = \tau_k(k) = 0 = \sigma(i)$.

Finally $k \mapsto \tau_k$ is injective because $\tau_k(0) = k$. $\square$

Theorem 4. *per $W_n = n + 1$ for every $n \geq 0$.*

Proof. By Lemma 2 the sum defining $\text{per } W_n$ counts the permutations satisfying (1), each contributing 1; by Lemma 3 there are $n + 1$ of them. $\square$

3 The conjecture

Corollary 5. *Every natural number m is the permanent of a square matrix with entries in $\{0, 1\}$.*

Proof. For $m = 0$ take the 1×1 zero matrix. For $m = k + 1$ take W_k and apply Theorem 4. $\square$

Corollary 6. *Conjecture 00000000050 is true.*

Proof. A prime is a natural number; apply Corollary 5. Explicitly, for a prime p the matrix W_{p-1} is a $p \times p$ matrix with entries in $\{0, 1\}$ and $\text{per } W_{p-1} = p$. $\square$

Remark 7. The hypothesis that p be prime is not used anywhere, and cannot be: the permanents of $\{0, 1\}$ matrices are exactly the natural numbers, by Corollary 5 in one direction and because a permanent of a nonnegative integer matrix is a nonnegative integer in the other. A conjecture restricted to primes is therefore a special case of a statement with no arithmetic content. Read as a question about the *smallest* matrix with a given permanent — the minimal N for which some $N \times N$ $\{0, 1\}$ matrix has permanent p — it would not be vacuous; our construction gives $N \leq p$, and we do not address minimality.

Remark 8. The construction has a transparent combinatorial reading: $\text{per } A$ counts the perfect matchings of the bipartite graph with biadjacency matrix A . For W_n , rows $1, \dots, n$ each see only columns 0 and their own, so a perfect matching either matches every $r \neq 0$ to column r — leaving row 0 for column 0 — or matches a single $r \neq 0$ to column 0, forcing row 0 to column r . That is $n + 1$ matchings.

4 Formal verification

The accompanying Lean 4 project formalizes the proof against Mathlib. The conjecture is stated in the formalization, so the final theorem is a formalization of the conjecture itself. Permanents are Mathlib's `Matrix.permanent`, not a local redefinition.

- `W`, `W_apply`, `W_zero_or_one` — the matrix and the fact that its entries lie in $\{0, 1\}$.

- `Contributes, prod_W` — Lemma 2: the product over a permutation is 1 or 0 according to (1).
- `contributes_swap, eq_swap_of_contributes, swap_injective, filter_contributes` — Lemma 3: the contributing permutations are exactly the $n + 1$ transpositions $(0\ k)$.
- `permanent_W` — Theorem 4.
- `exists_zeroOne_permanent` — Corollary 5.
- `ConjectureHolds, conjecture_00000000050_true` — the conjecture and its proof.

The project builds with `lake build` on `leanprover/lean4:v4.33.1` with `Mathlib v4.33.1`. It contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms of its own: `#print axioms conjecture_00000000050_true` reports exactly the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.